

“The Essentials of Computer Organization and Architecture”

Unit 6 – Exercises and Solutions

ES1 Find the CPU utilization when polling i) a mouse, ii) a floppy disk drive, and iii) a hard disk. [consider the information in (*)]

(*) complementary data:

- CPU Clock Frequency:

500 MHz = 500 Mcycles/second = $500 \cdot 10^6$ cycles/second

- Polling Time:

400 cycles/polling

- Mouse Polling Rate:

30 pollings/second

- Floppy Disk Transfer Rate:

50 Kbytes/s = $50 \cdot 10^3$ bytes/s, 16 bits at a time

- Hard Disk Transfer Rate:

4 Mbytes/s = $4 \cdot 10^6$ bytes/s, 4 words ($4 \cdot 32$ bit) at a time

ES1 Find the CPU utilization when polling i) a mouse, ii) a floppy disk drive, and iii) a hard disk. [consider the information in (*)]

i) CPU utilization when polling the mouse

Solution:

Assuming that mouse utilization is **100% (unrealistic)**

number of mouse pollings = Mouse Polling Rate = 30/sec

cycles per polling = Polling Time = 400 cycles/polling

number of cpu cycles = CPU Clock Frequency = 500Mcycles/sec

$$\begin{aligned}\text{CPU utilization} &= \frac{\text{number of mouse pollings} * \text{cycles per polling}}{\text{number of cpu cycles}} * 100\% \\ &= (30 * 400) / (500 * 10^6) \\ &= 12000 / (500 * 10^6) = \mathbf{0.0024\%}\end{aligned}$$

ES1 Find the CPU utilization when polling i) a mouse, ii) a floppy disk drive, and iii) a hard disk. [consider the information in (*)]

ii) CPU utilization when polling the floppy disk drive

Solution:

Assuming that FDD utilization is **100% (unrealistic)**

Floppy Disk Transfer Rate = 50 Kbytes/s

num. of bytes transferred per polling = 2 bytes (16 bits)

$$\begin{aligned} \text{num. of floppy disk pollings} &= \frac{\text{Floppy Disk Transfer Rate}}{\text{num. of bytes transferred per polling}} \\ &= (50 \times 10^3) / 2 = 25000 \text{ pollings/sec} \end{aligned}$$

cycles per polling = Polling Time = 400 cycles/polling

number of cpu cycles = CPU Clock Frequency = 500 Mcycles/sec

$$\begin{aligned} \text{CPU utilization} &= \frac{\text{num. of floppy disk pollings} \times \text{cycles per polling}}{\text{number of cpu cycles}} \times 100\% \\ &= (25000 \times 400) / (500 \times 10^6) \\ &= (10 \times 10^6) / (500 \times 10^6) = 2\% \end{aligned}$$

ES1 Find the CPU utilization when polling i) a mouse, ii) a floppy disk drive, and iii) a hard disk. [consider the information in (*)]

iii) CPU utilization when polling the hard drive

Solution:

Assuming that HDD utilization is **100% (unrealistic)**

Hard Disk Transfer Rate = 4 Mbytes/s = 4×10^6 bytes/s

num. of bytes transferred per polling = 16 bytes (4×32 bits)

$$\begin{aligned} \text{num. of hard disk pollings} &= \frac{\text{Hard Disk Transfer Rate}}{\text{num. of bytes transferred per polling}} \\ &= (4 \times 10^6) / 16 = 250000 \text{ pollings/sec} \end{aligned}$$

cycles per polling = Polling Time = 400 cycles/polling

number of cpu cycles = CPU Clock Frequency = 500Mcycles/sec

$$\begin{aligned} \text{CPU utilization} &= \frac{\text{num. of hard disk pollings} \times \text{cycles per polling}}{\text{number of cpu cycles}} \times 100\% \\ &= (250000 \times 400) / (500 \times 10^6) \\ &= (100 \times 10^6) / (500 \times 10^6) = 20\% \end{aligned}$$

ES2 Find the CPU utilization when using interrupts to interact with a hard disk. [consider the information in (*)]

(*) complementary data:

- CPU Clock Frequency:
 $500 \text{ MHz} = 500 \text{ Mcycles/second} = 500 * 10^6 \text{ cycles/second}$
- Interrupt Time:
 $500 \text{ cycles/interrupt}$
- Disk Utilization: **100%** → for a fair comparison with polling ...
- Hard Disk Transfer Rate:
 $4 \text{ Mbytes/s} = 4 * 10^6 \text{ bytes/s}$, 4 words ($4 * 32 \text{ bit}$) at a time

ES2 Find the CPU utilization when using interrupts to interact with a hard disk. [consider the information in (*)]

CPU utilization when using interrupts to interact with the Disk

Solution:

In this scenario, using interrupts is not better than using polling, because handling one interrupt requires 100 more cycles than one polling ... However, for Disk utilization bellow 100% (more realistic), using interrupts would largely pay off.

Hard Disk Transfer Rate = 4 Mbytes/s = $4 \cdot 10^6$ bytes/s

num. of bytes transferred per interrupt = 16 bytes ($4 \cdot 32$ bits)

$$\text{num. of hard disk interrupts} = \frac{\text{Hard Disk Transfer Rate}}{\text{num. of bytes transferred per interrupt}} \\ = (4 \cdot 10^6) / 16 = 250000 \text{ interrupts/sec}$$

cycles per interrupt = Interrupt Time = 500 cycles/interrupt

number of cpu cycles = CPU Clock Frequency = 500 Mcycles/sec

$$\text{CPU utilization} = \frac{\text{num. of hard disk interrupts} * \text{cycles per interrupt} * 100\%}{\text{number of cpu cycles}} \\ = (250000 \cdot 500) / (500 \cdot 10^6) \\ = (125 \cdot 10^6) / (500 \cdot 10^6) = 25\%$$

ES3 Find the CPU utilization when using DMA transfers between hard disk and RAM. [consider the information in (*)]

(*) complementary data:

- CPU Clock Frequency:
 $500 \text{ MHz} = 500 \text{ Mcycles/second} = 500 * 10^6 \text{ cycles/second}$
- “DMA Operation Start” Time:
1000 cycles
- “DMA Operation End” Interrupt Time
500 cycles
- DMA Operation Transfer:
8 Kbytes = $8 * 10^3 \text{ bytes}$
- Hard Disk Transfer Rate:
 $4 \text{ Mbytes/s} = 4 * 10^6 \text{ bytes/s}$
- Disk Utilization: **100%** →

for a fair comparison with
polling and interrupts ...

ES3 Find the CPU utilization when using DMA transfers between hard disk and RAM. [consider the information in (*)]

CPU utilization when using DMA transfers

Solution:

In this scenario, the CPU utilization (**0.15%**) is MUCH lower than the one with polling (**20%**) or interrupts (**25%**).

Hard Disk Transfer Rate = 4 Mbytes/s = $4 \cdot 10^6$ bytes/s

num. bytes transfer. per DMA = DMA Operation Transfer = $8 \cdot 10^3$ bytes

$$\begin{aligned} \text{num. DMA transfers} &= \frac{\text{Hard Disk Transfer Rate}}{\text{num. bytes transfer. per DMA}} \\ &= (4 \cdot 10^6) / 8 \cdot 10^3 = 500 \text{ transf./second} \end{aligned}$$

cycles per DMA = “DMA Operation Start” Time + “DMA Operation End”

Interrupt Time = 1000 + 500 = 1500 cycles/DMA

number of cpu cycles = CPU Clock Frequency = 500 Mcycles/sec

$$\begin{aligned} \text{CPU utilization} &= \frac{\text{num. DMA transfers} \cdot \text{cycles per DMA}}{\text{number of cpu cycles}} \cdot 100\% = \\ &= (500 \cdot 1500) / (500 \cdot 10^6) \\ &= (750 \cdot 10^3) / (500 \cdot 10^6) = 0.15\% \end{aligned}$$

ES4 A peripheral generates 80 MBytes of data every second, that must be collected by the host system. The host CPU has a clock frequency of 1 GHz. For each of the following techniques of I/O control, provide the requested values, without using powers, scientific notation or SI symbols (K, M, etc.) and respecting the precision required.

Note: in this question, the symbols K, M, etc. refer to powers of 10.

a) The CPU interacts with the peripheral through polling. Each polling operation consumes 500 CPU clock cycles and is capable of collecting 512 bytes of data from the peripheral.

a1) The number of pollings per second required to sustain the peripheral throughput is: _____

a2) The number of CPU cycles spent to make the pollings calculated in a1) is: _____

a3) The CPU occupation rate (in %) represented by the value calculated in a2) is : _____ % (value to the hundredths, truncated)

ES4 A peripheral generates 80 MBytes of data every second, that must be collected by the host system. The host CPU has a clock frequency of 1 GHz. For each of the following techniques of I/O control, provide the requested values, without using powers, scientific notation or SI symbols (K, M, etc.) and respecting the precision required.

a) The CPU interacts with the peripheral through polling. Each polling operation consumes 500 CPU clock cycles and is capable of collecting 512 bytes of data from the peripheral.

Problema Data:

A = peripheral transfer rate = 80 Mbytes / s = 80x10⁶ bytes/s

B = CPU clock frequency = 1 GHz

= number of CPU clock cycles per second = 1x10⁹ cycles/s

C = number of CPU cycles consumed per **polling** = 500 cycles

D = number of bytes transferred per **polling** = 512 bytes

Resolution:

E = number of **pollings** necessary to transfer 80 Mbytes in a second

E = A / D = 80 M / 512 = 80x10⁶ / 512 = **156250 pollings**

F = number of CPU cycles spent in E **pollings** = E x C = 156250 x 500 = **78125000 cycles**

G = CPU occupation rate = F / B = 78125000 / (1x10⁹) = **0.078125**

Answer to a1): 156250

Answer to a2): 78125000

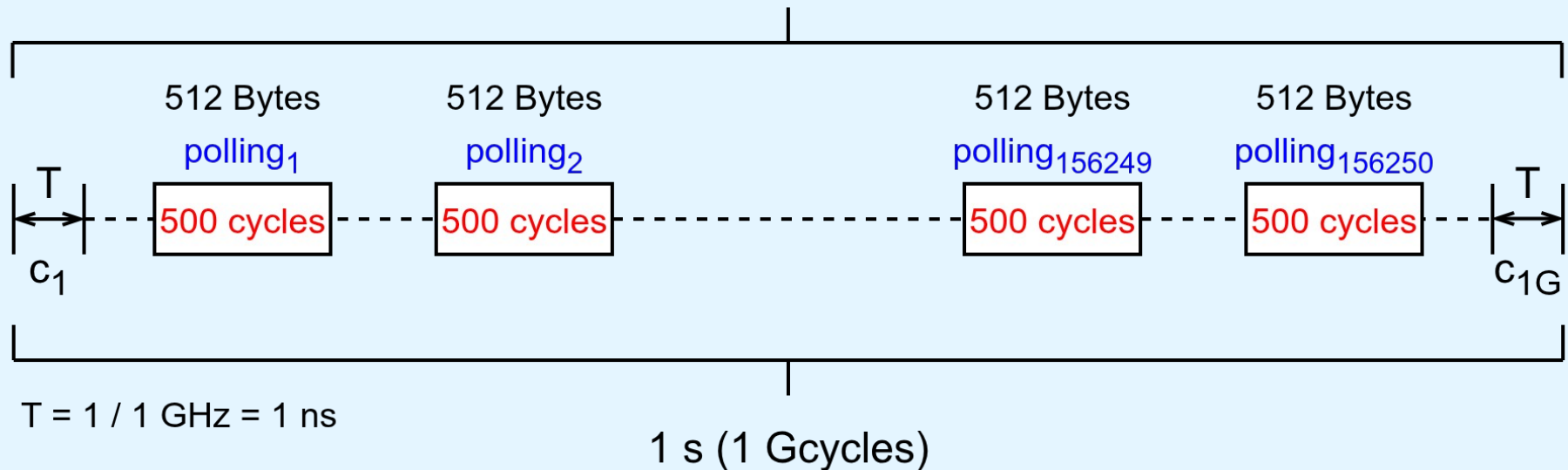
Answer to a3): 7.81 % (value to the hundredths, truncated)

ES4 A peripheral generates 80 MBytes of data every second, that must be collected by the host system. The host CPU has a clock frequency of 1 GHz. For each of the following techniques of I/O control, provide the requested values, without using powers, scientific notation or SI symbols (K, M, etc.) and respecting the precision required.

a) The CPU interacts with the peripheral through polling. Each polling operation consumes 500 CPU clock cycles and is capable of collecting 512 bytes of data from the peripheral.

Scenario:

80 MBytes (156250 pollings) (78125000 cycles) (7,81% CPU occupancy)



ES4 A peripheral generates 80 MBytes of data every second, that must be collected by the host system. The host CPU has a clock frequency of 1 GHz. For each of the following techniques of I/O control, provide the requested values, without using powers, scientific notation or SI symbols (K, M, etc.) and respecting the precision required.

Note: in this question, the symbols K, M, etc. refer to powers of 10.

b) The CPU interacts with the peripheral through interrupts generated by the peripheral. Each interrupt handling consumes 1000 CPU clock cycles and is capable of collecting 2K bytes of data from the peripheral.

b1) The number of interrupts per second required to sustain the peripheral throughput is: _____

b2) The number of CPU cycles spent to handle the interrupts calculated in b1) is: _____

b3) The CPU occupation rate (in %) represented by the value calculated in b2) is : _____ % (value to the hundredths, truncated)

ES4 A peripheral generates 80 MBytes of data every second, that must be collected by the host system. The host CPU has a clock frequency of 1 GHz. For each of the following techniques of I/O control, provide the requested values, without using powers, scientific notation or SI symbols (K, M, etc.) and respecting the precision required.

b) The CPU interacts with the peripheral through interrupts generated by the peripheral. Each interrupt handling consumes 1000 CPU clock cycles and is capable of collecting 2K bytes of data from the peripheral.

Problema Data:

A = peripheral transfer rate = 80 Mbytes / s = 80×10^6 bytes/s

B = CPU clock frequency = 1 GHz

= number of CPU clock cycles per second = 1×10^9 cycles/s

C = number of CPU cycles consumed per interrupt = 1000 cycles

D = number of bytes transferred per interrupt = 2 Kbytes = 2×10^3 bytes

Resolution:

E = number of interrupts necessary to transfer 80 Mbytes in a second

$E = A / D = 80 \text{ M} / 2 \text{ K} = 80 \times 10^6 / 2 \times 10^3 = 40000$ interrupts

F = number of CPU cycles spent in E interrupts = $E \times C = 40000 \times 1000 = 40000000$ cycles

G = CPU occupation rate = $F / B = (40 \times 10^6) / (1 \times 10^9) = 0.04$

Answer to b1): 40000

Answer to b2): 40000000

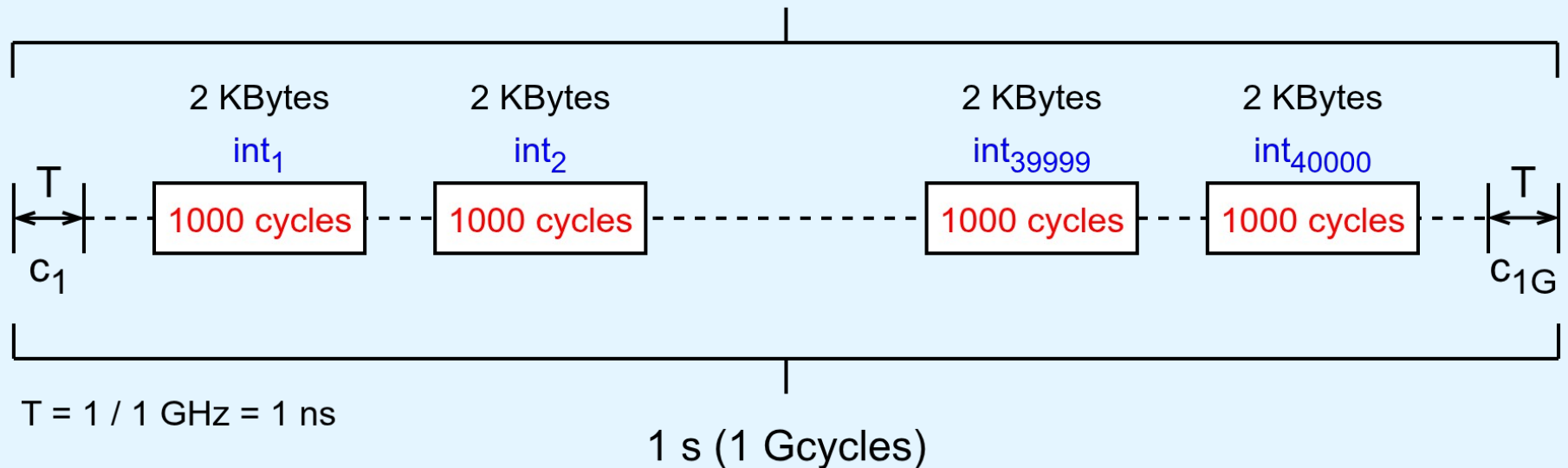
Answer to b3): 4.00 % (value to the hundredths, truncated)

ES4 A peripheral generates 80 MBytes of data every second, that must be collected by the host system. The host CPU has a clock frequency of 1 GHz. For each of the following techniques of I/O control, provide the requested values, without using powers, scientific notation or SI symbols (K, M, etc.) and respecting the precision required.

b) The CPU interacts with the peripheral through interrupts generated by the peripheral. Each interrupt handling consumes 1000 CPU clock cycles and is capable of collecting 2K bytes of data from the peripheral.

Scenario:

80 MBytes (40000 interrupts) (40000000 cycles) (4% CPU occupancy)



ES4 A peripheral generates 80 MBytes of data every second, that must be collected by the host system. The host CPU has a clock frequency of 1 GHz. For each of the following techniques of I/O control, provide the requested values, without using powers, scientific notation or SI symbols (K, M, etc.) and respecting the precision required.

Note: in this question, the symbols K, M, etc. refer to powers of 10.

c) The CPU uses a DMA controller to manage the interaction with the peripheral. The start and end of a DMA operation consumes 2000 and 1000 CPU clock cycles respectively. Each DMA operation collects 128 KBytes of data from the peripheral.

c1) The number of DMA operations per second required to sustain the peripheral throughput is: _____

c2) The number of CPU cycles spent to start and end DMA operations calculated in c1) is: _____

c3) The CPU occupation rate (in %) represented by the value calculated in c2) is : _____ % (value to the hundredths, truncated)

ES4 A peripheral generates 80 MBytes of data every second, that must be collected by the host system. The host CPU has a clock frequency of 1 GHz. For each of the following techniques of I/O control, provide the requested values, without using powers, scientific notation or SI symbols (K, M, etc.) and respecting the precision required.

c) The CPU uses a DMA controller to manage the interaction with the peripheral. The start and end of a DMA operation consumes 2000 and 1000 CPU clock cycles respectively. Each DMA operation collects 128 KBytes of data from the peripheral.

Problema Data:

A = peripheral transfer rate = 80 Mbytes / s = 80×10^6 bytes/s

B = CPU clock frequency = 1 GHz

= number of CPU clock cycles per second = 1×10^9 cycles/s

Cs = number of CPU cycles consumed to **start a DMA operation** = 2000 cycles

Ce = number of CPU cycles consumed to **end a DMA operation** = 1000 cycles

D = number of bytes transferred per **DMA operation** = 128 Kbytes = 128×10^3 bytes

Resolution:

E = number of **DMA operations** necessary to transfer 80 Mbytes in a second

$E = A / D = 80 \text{ M} / 128 \text{ K} = 80 \times 10^6 / 128 \times 10^3 = \mathbf{625 \text{ DMA operations}}$

F = number of CPU cycles spent in E **DMA operations** = $E \times (Cs + Ce) = 625 \times 3000 = \mathbf{1875000 \text{ cycles}}$

G = CPU occupation rate = $F / B = (1875000) / (1 \times 10^9) = \mathbf{0.001875}$

Answer to c1): 625

Answer to c2): 1875000

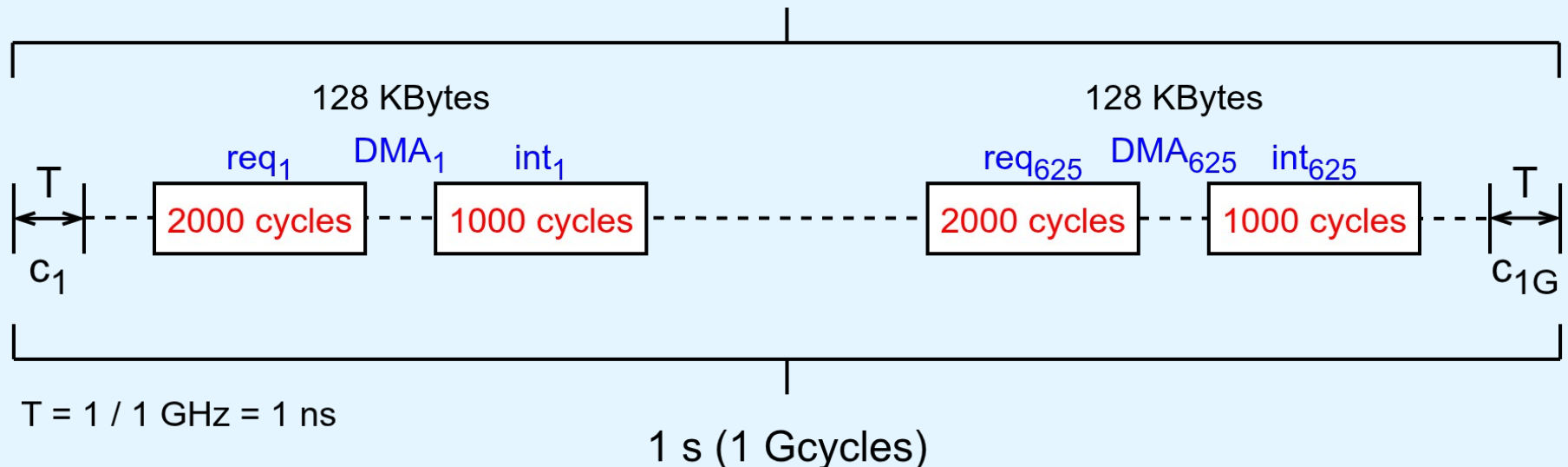
Answer to c3): 0.18 % (value to the hundredths, truncated)

ES4 A peripheral generates 80 MBytes of data every second, that must be collected by the host system. The host CPU has a clock frequency of 1 GHz. For each of the following techniques of I/O control, provide the requested values, without using powers, scientific notation or SI symbols (K, M, etc.) and respecting the precision required.

c) The CPU uses a DMA controller to manage the interaction with the peripheral. The start and end of a DMA operation consumes 2000 and 1000 CPU clock cycles respectively. Each DMA operation collects 128 KBytes of data from the peripheral.

Scenario:

80 MBytes (625 DMAs) (1875000 cycles) (0,18% CPU occupancy)



HD1 A non-ZBR hard disk has 4 surfaces, 1024 tracks per surface, 128 sectors per track, 512 bytes per sector, a seek time of 5 milliseconds, and a rotational speed of 5000 RPM.

a) What is the capacity of the drive (in MBytes) ?

b) What is the access time (in ms) ?

HD1 A non-ZBR hard disk has 4 surfaces, 1024 tracks per surface, 128 sectors per track, 512 bytes per sector, a seek time of 5 milliseconds, and a rotational speed of 5000 RPM.

a) What is the capacity of the drive (in MBytes) ?

Solution:

$$\begin{aligned} &4 \text{ surfaces} \times \\ &1024 \text{ tracks/surface} \times \\ &128 \text{ sectors/track} \times \\ &512 \text{ bytes/sector} = 4 \times 1024 \times 128 \times 512 = 268435456 \text{ bytes} \\ &= 2^2 \times 2^{10} \times 2^7 \times 2^9 = \mathbf{2^{28} \text{ bytes}} \\ &= 2^{28} / \mathbf{10^6} \text{ MBytes} = \mathbf{268,43 \text{ MBytes}} \end{aligned}$$

Hard drive capacity is
measured in powers of 10 !

HD1 A non-ZBR hard disk has 4 surfaces, 1024 tracks per surface, 128 sectors per track, 512 bytes per sector, a seek time of 5 milliseconds, and a rotational speed of 5000 RPM.

b) What is the access time (in ms) ?

Solution:

rotational latency:

5000 rot ----- 60 s

1 rot ----- x s

$x = 60 / 5000 = 0.012 \text{ s}$

$\bar{x} = x / 2 = 0.006 \text{ s} = \mathbf{6 \text{ ms}}$

seek time = 5 ms

rotational latency = $[(60 \text{ s} / 5000 \text{ rpm}) \times (1000 \text{ ms} / \text{s})] / 2 = \mathbf{6 \text{ ms}}$

access time = 5 ms + 6 ms = 11 ms

HD2 A non-ZBR hard disk has 10 cylinders, 4 heads, 100 sectors per track, and 512 bytes per sector.

- a) What is the capacity of the drive (in MBytes) ?
- b) The LBA index of a sector is 1234. Compute the CHS (Cylinder, Head, Sector) coordinates of that sector.

$$\text{LBA} = (C \times \text{HPC} + H) \times \text{SPT} + (S - 1)$$

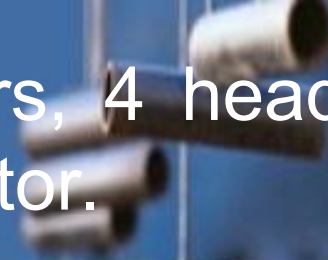
$$C = \text{LBA} \div (\text{HPC} \times \text{SPT})$$

$$H = (\text{LBA} \div \text{SPT}) \bmod \text{HPC}$$

$$S = (\text{LBA} \bmod \text{SPT}) + 1$$

HPC = Heads Per Cylinder (Surfaces)

SPT = Sectors Per Track



HD2 A non-ZBR hard disk has 10 cylinders, 4 heads, 100 sectors per track, and 512 bytes per sector.

a) What is the capacity of the drive (in MBytes) ?

Solution:

4 heads => 4 surfaces

10 cylinders => 10 tracks per surface

capacity of the drive = 4 surfaces x 10 tracks per surface x
100 sectors per track x 512 bytes per sector
= 4000 x 512 = 2048000 bytes
= 2048000 / 10⁶ MBytes = **2,048 MBytes**

HD2 A non-ZBR hard disk has 10 cylinders, 4 heads, 100 sectors per track, and 512 bytes per sector.

b) The LBA index of a sector is 1234. Compute the CHS (Cylinder,Head,Sector) coordinates of that sector.

Solution:

$$\text{LBA} = (C \times \text{HPC} + H) \times \text{SPT} + (S - 1)$$

$$C = \text{LBA} \div (\text{HPC} \times \text{SPT})$$

$$H = (\text{LBA} \div \text{SPT}) \bmod \text{HPC}$$

$$S = (\text{LBA} \bmod \text{SPT}) + 1$$

HPC = Heads Per Cylinder (Surfaces)

SPT = Sectors Per Track

$$\text{HPC} = 4; \text{SPT} = 100$$

$$C = 1234 \div (4 \times 100) = 1234 \div 400 = 3$$

$$H = (1234 \div 100) \bmod 4 = 12 \bmod 4 = 0$$

$$S = (1234 \bmod 100) + 1 = 34 + 1 = 35$$

$$\text{confirmation: LBA} = (3 \times 4 + 0) \times 100 + (35 - 1) = 12 \times 100 + 34 = 1234$$

HD3 A non-ZBR hard disk has 20 cylinders, 6 heads, 1800 sectors per track, and 4096 bytes per sector.

- a) What is the capacity of the drive (in GBytes) ?
- b) The LBA index of a sector is 123456. Compute the CHS (Cylinder, Head, Sector) coordinates of that sector.

$$\text{LBA} = (C \times \text{HPC} + H) \times \text{SPT} + (S - 1)$$

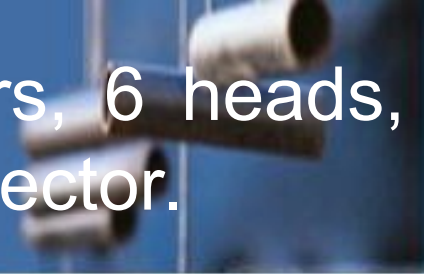
$$C = \text{LBA} \div (\text{HPC} \times \text{SPT})$$

$$H = (\text{LBA} \div \text{SPT}) \bmod \text{HPC}$$

$$S = (\text{LBA} \bmod \text{SPT}) + 1$$

HPC = Heads Per Cylinder (Surfaces)

SPT = Sectors Per Track



HD3 A non-ZBR hard disk has 20 cylinders, 6 heads, 1800 sectors per track, and 4096 bytes per sector.

a) What is the capacity of the drive (in GBytes) ?

Solution:

6 heads => 6 surfaces

20 cylinders => 20 tracks per surface

disk capacity = 6 surfaces x 20 tracks per surface x
1800 sectors per track x 4096 bytes/sector
= 884736000 bytes
= 884736000/10⁹ GBytes = **0,884736 GBytes**

HD3 A non-ZBR hard disk has 20 cylinders, 6 heads, 1800 sectors per track, and 4096 bytes per sector.

b) The LBA index of a sector is 123456. Compute the CHS (Cylinder, Head, Sector) coordinates of that sector.

Solution:

$$\text{LBA} = (C \times \text{HPC} + H) \times \text{SPT} + (S - 1)$$

$$C = \text{LBA} \div (\text{HPC} \times \text{SPT})$$

$$H = (\text{LBA} \div \text{SPT}) \bmod \text{HPC}$$

$$S = (\text{LBA} \bmod \text{SPT}) + 1$$

HPC = Heads Per Cylinder (Surfaces)

SPT = Sectors Per Track

$$\text{HPC} = 6; \text{SPT} = 1800$$

$$C = 123456 \div (6 \times 1800) = 123456 \div 10800 = 11$$

$$H = (123456 \div 1800) \bmod 6 = 68 \bmod 6 = 2$$

$$S = (123456 \bmod 1800) + 1 = 1056 + 1 = 1057$$

$$\text{confirmation: LBA} = (11 \times 6 + 2) \times 1800 + (1057 - 1) = 68 \times 1800 + 1056 = 123456$$

HD4 Consider a non-ZBR hard disk consisting of X platters, with all surfaces used. Each surface is organized into Y tracks, and each track comprises Z sectors (of W bytes each). It is also known that the disk rotates R times per minute, and that the time it takes a read/write head to position itself on top of the intended track is T seconds.

Let $X=3$, $Y=100$, $Z=1000$, $W=512$, $R=4900$ e $T=0.005$.

a) Indicate the total disk capacity, expressed in MBytes.

b1) Indicate the time (in milliseconds) it takes, on average, to start reading the first desired sector of a certain track, assuming that the read/write head is already positioned over that track.

b2) Solve b1) again, now assuming that the read/write head is not yet positioned over the correct track.

c) Indicate the number of read/write heads and cylinders on this hard drive.

d) Indicate the CHS coordinates equivalent to the coordinate LBA=12345.

LBA = $(C \times \text{HPC} + H) \times \text{SPT} + (S - 1)$; **C** = $\text{LBA} \div (\text{HPC} \times \text{SPT})$

H = $(\text{LBA} \div \text{SPT}) \bmod \text{HPC}$; **S** = $(\text{LBA} \bmod \text{SPT}) + 1$

HPC = Heads Per Cylinder (Surfaces) ; **SPT** = Sectors Per Track

HD4 Consider a non-ZBR hard disk consisting of X platters, with all surfaces used. Each surface is organized into Y tracks, and each track comprises Z sectors (of W bytes each). It is also known that the disk rotates R times per minute, and that the time it takes a read/write head to position itself on top of the intended track is T seconds.

Let $X=3$, $Y=100$, $Z=1000$, $W=512$, $R=4900$ e $T=0.005$.

a) Indicate the total disk capacity, expressed in MBytes.

Resolution:

disk capacity = X platters * 2 surfaces per platter * Y tracks per surface * Z sectors per track * W bytes per sector = $3 * 2 * 100 * 1000 * 512 = 307200000$ bytes = $307200000 / 10^6$ Mbytes = **307.2** MBytes

Answer: 307.20 MBytes (value to hundredths, truncated)

HD4 Consider a non-ZBR hard disk consisting of X platters, with all surfaces used. Each surface is organized into Y tracks, and each track comprises Z sectors (of W bytes each). It is also known that the disk rotates R times per minute, and that the time it takes a read/write head to position itself on top of the intended track is T seconds.

Let $X=3$, $Y=100$, $Z=1000$, $W=512$, $R=4900$ e $T=0.005$.

b1) Indicate the time (in milliseconds) it takes, on average, to start reading the first desired sector of a certain track (assuming that the read/write head is already positioned over that track).

Resolution:

The time requested is nothing else than the (average) rotational delay, which is half of the time needed for a disk rotation (turn).

Time of one rotation = $60\text{s} / R \text{ rotations} = 60 / 4900 = 0.012244... \text{ s}$.

Time of half a rotation = $0.012244... / 2 = 0.006122... \text{ s} = 6.122... \text{ ms}$

Answer: 6 milliseconds (integer part)

HD4 Consider a non-ZBR hard disk consisting of X platters, with all surfaces used. Each surface is organized into Y tracks, and each track comprises Z sectors (of W bytes each). It is also known that the disk rotates R times per minute, and that the time it takes a read/write head to position itself on top of the intended track is T seconds.

Let $X=3$, $Y=100$, $Z=1000$, $W=512$, $R=4900$ e $T=0.005$.

b2) Solve b1) again, now assuming that the read/write head is not yet positioned over the correct track.

Resolution:

In addition to the rotational latency (see b1)), it is also necessary to include the positioning time (seek time). In this way, the (average) access time becomes
 $T \text{ (seek time)} + 0.006 \text{ (rotational latency)} = 0.005 + 0.006122 = 0.011122 \dots \text{ s}$

Answer: 11 milliseconds (integer part)

HD4 Consider a non-ZBR hard disk consisting of X platters, with all surfaces used. Each surface is organized into Y tracks, and each track comprises Z sectors (of W bytes each). It is also known that the disk rotates R times per minute, and that the time it takes a read/write head to position itself on top of the intended track is T seconds.

Let $X=3$, $Y=100$, $Z=1000$, $W=512$, $R=4900$ e $T=0.005$.

c) Indicate the number of read/write heads and cylinders on this hard drive.

Resolution:

One head is required for each surface. If there are X platters, "with all surfaces used", then there are $X * 2 = 3 * 2 = 6$ surfaces, and so there are **6** heads.

There are as many cylinders as there are tracks per surface. Therefore, with $Y=100$ tracks per surface, there are **100** cylinders.

Answer: **6** heads ; **100** cylinders

HD4 Consider a non-ZBR hard disk consisting of X platters, with all surfaces used. Each surface is organized into Y tracks, and each track comprises Z sectors (of W bytes each). It is also known that the disk rotates R times per minute, and that the time it takes a read/write head to position itself on top of the intended track is T seconds.

Let $X=3$, $Y=100$, $Z=1000$, $W=512$, $R=4900$ e $T=0.005$.

d) Indicate the CHS coordinates equivalent to the coordinate $LBA=12345$.

$$\begin{aligned} \text{LBA} &= (C \times \text{HPC} + H) \times \text{SPT} + (S - 1) ; \text{C} = \text{LBA} \div (\text{HPC} \times \text{SPT}) \\ \text{H} &= (\text{LBA} \div \text{SPT}) \bmod \text{HPC} ; \text{S} = (\text{LBA} \bmod \text{SPT}) + 1 \\ \text{HPC} &= \text{Heads Per Cylinder (Surfaces)} ; \text{SPT} = \text{Sectors Per Track} \end{aligned}$$

Resolution:

$\text{HPC} = \text{number of surfaces} = X * 2 = 3 * 2 = 6$ (see c)) ; $\text{SPT} = Z = 1000$

$C = 12345 \div (6 * 1000) = 12345 \div 6000 = 2$

$H = (12345 \div 1000) \bmod 6 = 12 \bmod 6 = 0$

$S = (12345 \bmod 1000) + 1 = 345 + 1 = 346$

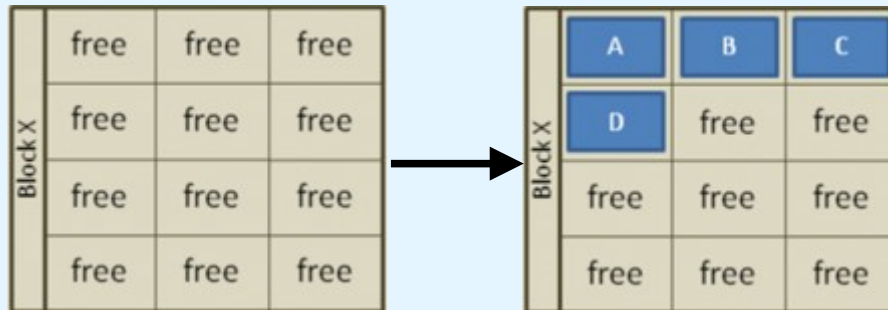
Confirmation:

$\text{LBA} = (2 * 6 + 0) * 1000 + (346 - 1) = 12 * 1000 + 345 = 12000 + 345 = 12345$

Answer: $C = \underline{2}$; $H = \underline{0}$; $S = \underline{346}$

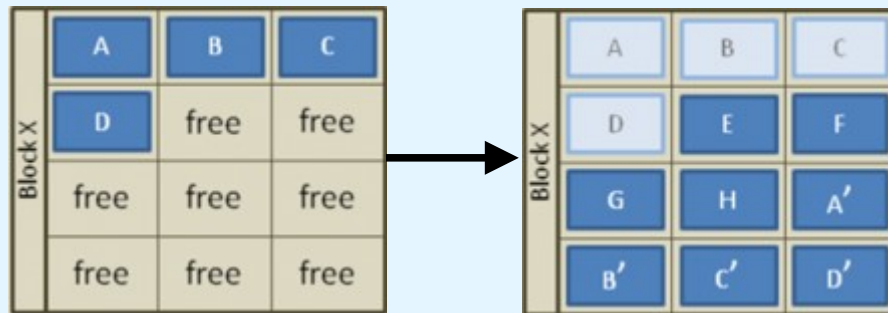
SSD1 Consider and SSD in which: i) reading a page takes 25 us, ii) writing into an empty page takes 250 us, iii) erasing a block takes 1500 us. Compute the time to perform the next operations:

a)



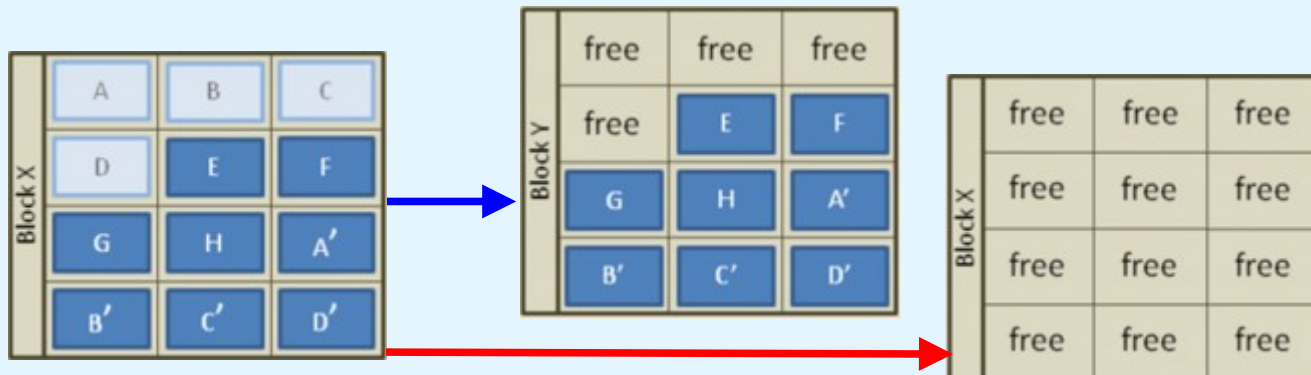
1. Four pages (A-D) are written to a block (X). Individual pages can be written at any time if they are currently free (erased).

b)



2. Four new pages (E-H) and four replacement pages (A'-D') are written to the block (X). The original A-D pages are now invalid (stale) data, but cannot be overwritten until the whole block is erased.

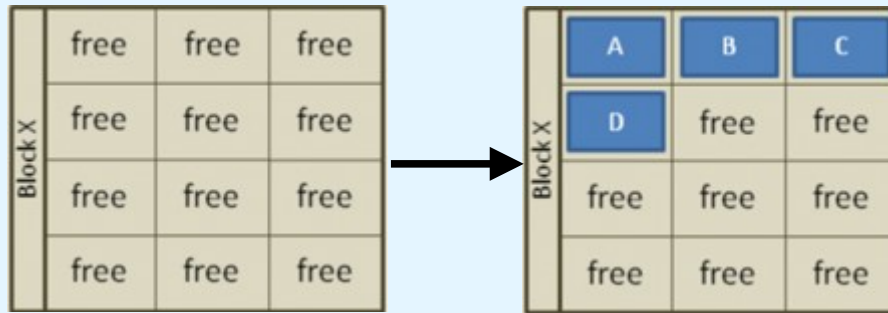
c)



3. In order to write to the pages with stale data (A-D) all good pages (E-H & A'-D') are read and written to a new block (Y) then the old block (X) is erased. This last step is garbage collection.

SSD1 Consider and SSD in which: i) reading a page takes 25 us, ii) writing into an empty page takes 250 us, iii) erasing a block takes 1500 us. Compute the time to perform the next operations:

a)

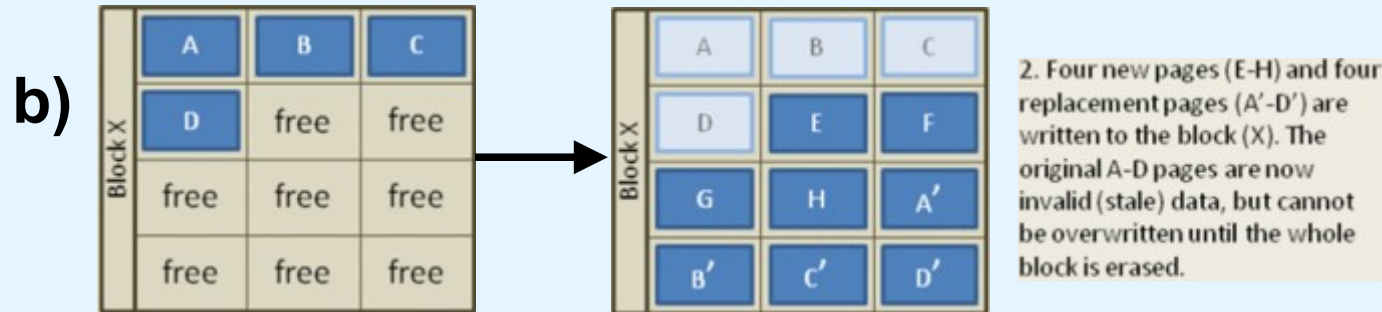


1. Four pages (A-D) are written to a block (X). Individual pages can be written at any time if they are currently free (erased).

Solution:

$$T = 4 \times 250 \text{ us (write A,B,C,D)} = 1000 \text{ us} = 1 \text{ ms}$$

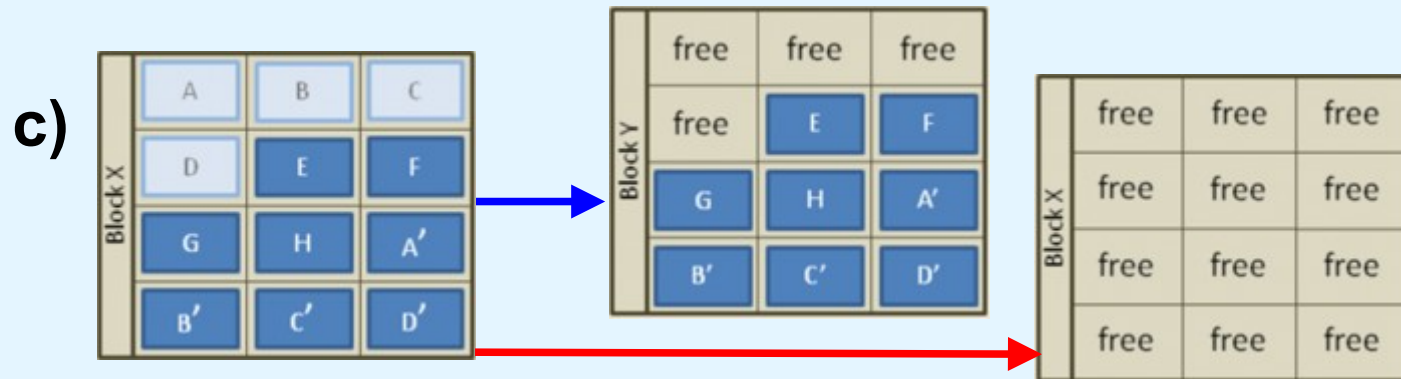
SSD1 Consider and SSD in which: i) reading a page takes 25 us, ii) writing into an empty page takes 250 us, iii) erasing a block takes 1500 us. Compute the time to perform the next operations:



Solution:

$$\begin{aligned}
 T &= 4 \times 250 \text{ us (write E,F,G,H)} + \\
 &4 \times 25 \text{ us (read A,B,C,D)} + \\
 &4 \times 250 \text{ us (write A',B',C',D')} = \\
 &1000 \text{ us} + 100 \text{ us} + 1000 \text{ us} = 2100 \text{ us} = 2,1 \text{ ms}
 \end{aligned}$$

SSD1 Consider and SSD in which: i) reading a page takes 25 us, ii) writing into an empty page takes 250 us, iii) erasing a block takes 1500 us. Compute the time to perform the next operations:



3. In order to write to the pages with stale data (A-D) all good pages (E-H & A'-D') are read and written to a new block (Y) then the old block (X) is erased. This last step is garbage collection.

Solution:

$$\begin{aligned}
 T &= 8 \times 25 \text{ us} \text{ (read E, F, ..., D' from block X)} + \\
 &\quad 8 \times 250 \text{ us} \text{ (write E, F, ..., D' in block Y)} + \\
 &\quad 1 \times 1500 \text{ us} \text{ (erase block X)} = \\
 &\quad 200 \text{ us} + 2000 \text{ us} + 1500 \text{ us} = 3700 \text{ us} = 3,7 \text{ ms}
 \end{aligned}$$

SSD2 On a 256 GB SSD with 16-page blocks and 16 KB pages, invalidating a page takes 1 μ s, reading a page takes 10 μ s, writing to an empty page takes 100 μ s, and erasing a block takes 1000 μ s, and all these operations are sequential, i.e. they are performed one page (or one block, as the case may be) at a time. The SSD is **empty**.

- a) State the storage capacity of each block, in KBytes.**
- b) State the total number of blocks on the SSD.**
- c) State the total number of pages on the SSD.**
- d) What is the minimum number of blocks needed to store a 1033 KBytes file ? (assume $K = 2^{10}$)**
- e) How many pages would be used under the conditions described in d)?**
- f) How many pages would remain free in the last block used by the file in d)?**
- g) How long (in microseconds) would it take to write the file mentioned in d)?**
- h) If, immediately after writing the file mentioned in d), it is necessary to modify 4 pages of that file, and the new version of those pages is created in the last block of the file, how long would that operation take?**
- i) If the 4 pages modified in h) were all in the same block B_x (where they were marked as invalid, after being copied to another block B_y), how much time would be spent on the garbage collection operation for block B_x (i.e., copying the valid pages of B_x to B_y and deleting B_x)?**

SSD2 On a 256 GB SSD with 16-page blocks and 16 KB pages, invalidating a page takes 1 μ s, reading a page takes 10 μ s, writing to an empty page takes 100 μ s, and erasing a block takes 1000 μ s, and all these operations are sequential, i.e. they are performed one page (or one block, as the case may be) at a time. The SSD is **empty**.

Solution:

a) State the storage capacity of each block, in KBytes.

$$\#BYTES(block) = \#PAGES(block) * \#BYTES(page) = 16 * 16 \text{ KBytes} = 256 \text{ KBytes}$$

b) State the total number of blocks on the SSD.

$$\begin{aligned} \#BLOCKS(SSD) &= \#BYTES(SSD) / \#BYTES(block) = \text{ceil} (256G / 256K) \\ &= \text{ceil} (256 * (10^9) / 256 * (2^{10})) = 976563 \text{ blocks} \end{aligned}$$

c) State the total number of pages on the SSD.

$$\#PAGES(SSD) = \#BLOCKS(SSD) * \#PAGES(block) = 976563 * 16 = 15625008 \text{ pages}$$

SSD2 On a 256 GB SSD with 16-page blocks and 16 KB pages, invalidating a page takes 1 μ s, reading a page takes 10 μ s, writing to an empty page takes 100 μ s, and erasing a block takes 1000 μ s, and all these operations are sequential, i.e. they are performed one page (or one block, as the case may be) at a time. The SSD is **empty**.

Solution (cont.):

**d) What's the minimum number of blocks needed to store a 1033 KBytes file ?
(assume $K=2^{10}$)**

- minimum number of blocks => use all pages of a free block before using another block
- $\#BLOCKS(file) = \text{ceil}(\#BYTES(file) / \#BYTES(block)) = \text{ceil}(1033K / 256K)$
 $= \text{ceil}(4,03...) = 5 \text{ blocks (4 fully used, 1 partially used)}$

e) How many pages would be used under the conditions described in d)?

$$\begin{aligned}\#PAGES(file) &= \text{ceil}(\#BYTES(file) / \#BYTES(page)) = \text{ceil}(1033K / 16K) = \\ &= \text{ceil}(64,56...) = 65 \text{ pages (16 + 16 + 16 + 16 + 1)}\end{aligned}$$

f) How many pages would remain free in the last block used by the file in d)?

$$\begin{aligned}\#PAGES(free) &= \#BLOCKS(file) * \#PAGES(block) - \#PAGES(file) \\ &= 5 * 16 - 65 = 15 \text{ pages}\end{aligned}$$

or

$$\begin{aligned}\#PAGES(free) &= \#PAGES(block) - (\#PAGES(file) \bmod \#PAGES(block)) \\ &= 16 - (65 \bmod 16) = 15 \text{ pages}\end{aligned}$$

SSD2 On a 256 GB SSD with 16-page blocks and 16 KB pages, invalidating a page takes 1 μ s, reading a page takes 10 μ s, writing to an empty page takes 100 μ s, and erasing a block takes 1000 μ s, and all these operations are sequential, i.e. they are performed one page (or one block, as the case may be) at a time. The SSD is **empty**.

Solution (cont.):

g) How long (in microseconds) would it take to write the file mentioned in d)?

- all blocks are initially free (the SSD is initially **empty**)
- 65 writes = $65 \cdot 100 \text{ us} = 6500 \text{ us}$

h) If, immediately after writing the file mentioned in d), it is necessary to modify 4 pages of that file, and the new version of those pages is created in the last block of the file, how long would that operation take?

$$4 \text{ reads} + 4 \text{ invalidations} + 4 \text{ writes} = 4 \cdot (10 + 1 + 100) = 4 \cdot 111 = 444 \text{ us}$$

i) If the 4 pages modified in h) were all in the same block Bx (where they were marked as invalid, after being copied to another block By), how much time would be spent on the garbage collection operation for block Bx (i.e., copying the valid pages of Bx to By and deleting Bx)?

- it is necessary to copy $16 - 4 = 12$ valid pages from that block, to another block
- $12 \text{ reads} + 12 \text{ writes} + 1 \text{ erasure} = 12 \cdot 10 \text{ us} + 12 \cdot 100 \text{ us} + 1 \cdot 1000 \text{ us} = 2320 \text{ us}$